MAIS PAC

IPhO 02.07: Magnetic Fields and Induction · Lecture + Problems

Magnetic Fields and Induction

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

7 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. Magnetic fields and the force on a moving charge

A magnetic field is the part of the electromagnetic field that acts on **moving** charge. Its strength and direction at a point are written B and measured in tesla, 1 {T} = 1 {N,s,C^(-1),m^(-1)}.

The force on a charge q moving with velocity v is

$$F = q, v\times B. \tag{1}$$

Read that formula slowly, because all of magnetism's odd behaviour comes out of it.

- The force is **perpendicular to both** v and B. With your right hand: fingers along v, curl towards B, thumb points along v× B for a positive charge (and opposite for a negative one).
- The force is **proportional to the sine** of the angle between them: F = qvBsin θ. It vanishes when the charge moves **along** the field and is maximal when it moves **across** it.

- Because the force is perpendicular to the velocity, it can never change the speed. A magnetic field does **no work** on a free charge, and the kinetic energy is conserved. This single sentence solves a large fraction of olympiad magnetism questions.
- The force is proportional to v, so a charge at rest feels nothing at all. There is no "magnetic charge" to feel it.

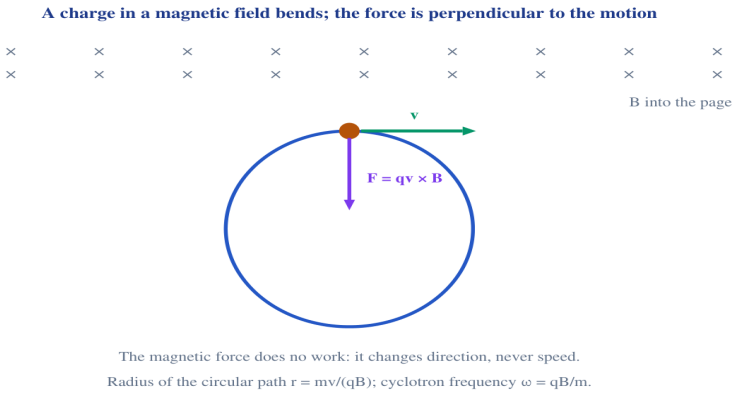


Figure 2. Because the force is always perpendicular to v, the magnetic field cannot change a particle's kinetic energy — a fact that solves many olympiad questions instantly.

A common mistake is to think the field energy comes from the particle. It does not: the field changes the momentum direction, and the energy bookkeeping stays flat.

2. Fields produced by currents: Biot–Savart and Ampère

A current is moving charge, so it produces a magnetic field. The key results, all worth memorising:

$$\{\text{long straight wire: }\}\ B = (\mu_0 I)/(2\pi r),\ \{\text{centre of a loop: }\}\ B = (\mu_0 I)/(2R),\ \{\text{long solenoid: }\}\ B = \mu_0 n I, \tag{2}$$

with μ_0 = 4π×10^(-7) {T,m,A^(-1)} and n the turns per metre. The direction is given by the right-hand rule: grasp the wire with your right hand, thumb along the current, and the fingers curl the way the field lines circle.

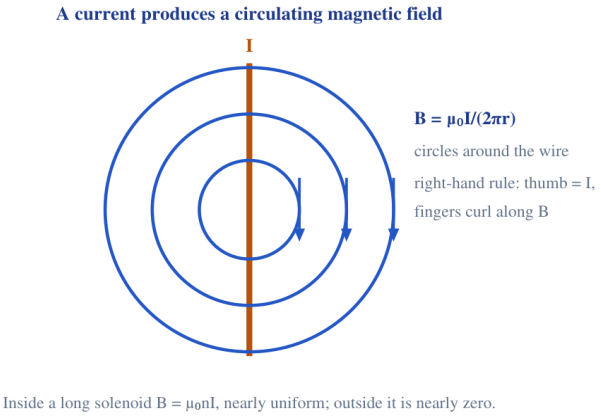


Figure 1. The field lines are closed loops with no start or end: there are no magnetic monopoles, which is why ∇·B = 0.

The general law behind these is the **Biot–Savart law**,

$$d B = (μ_0)/(4π),(I,d×{ r})/(r²), \quad (3)$$

and the integral form is **Ampère's law**,

$$B· d = μ_0 I_({enclosed}). \quad (4)$$

Ampère's law plays the same role for magnetism that Gauss's law plays for electrostatics: always true, only useful when symmetry makes B constant along a chosen loop. It is also incomplete — Maxwell added the displacement current so that a charging capacitor's gap still carries a field — but for steady currents the form above is enough.

The field lines of B are always **closed loops**: there are no magnetic monopoles. That is why the flux of B through any closed surface is zero, B· d A = 0, whereas the flux of E gives the enclosed charge.

3. Charged particles in magnetic fields: circular motion

If a charge moves perpendicular to a uniform field, the constant perpendicular force makes it travel in a circle:

$$qvB = (mv²)/(r) \; r = (mv)/(qB) = (p)/(qB). \quad (5)$$

The radius grows with momentum and shrinks with field. The period, however,

$$T = (2π \; r)/(v) = (2π \; m)/(qB), \; ω = (qB)/(m), \quad (6)$$

does **not** depend on the speed. That is the cyclotron principle: particles of the same species always come round in step, so an alternating voltage of the right frequency can accelerate them again and again. The kinetic energy grows without limit while the period stays fixed — until relativity spoils the trick and the mass effectively rises.

Interactive experiment: Circular motion in a magnetic field — open the online handout to use it.

If the velocity has a component along the field, that component is untouched while the perpendicular part circles: the path is a **helix** of pitch v_() T. This is how a magnetic field traps particles in the van Allen belts and confines plasma in a tokamak, and it is the principle of the mass spectrometer: for a given speed or given accelerating voltage, the radius depends on m/q, so ions of different isotopes land at different places.

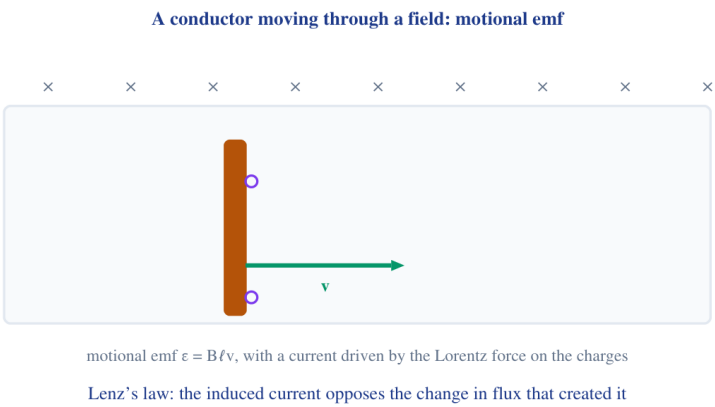


Figure 3. The rod experiences a retarding force BI■, so pushing it at constant speed requires mechanical power exactly equal to the electrical power I²R.

4. Force on a current-carrying conductor

A wire carrying current I in a field feels a force because the drifting charges inside it do. For a straight length ,

$$F = I,\times B,\ F = BIsin\ \theta. \tag{7}$$

Two parallel wires each sit in the other's field, so they attract when the currents are in the same direction and repel when they are opposite, with force per unit length

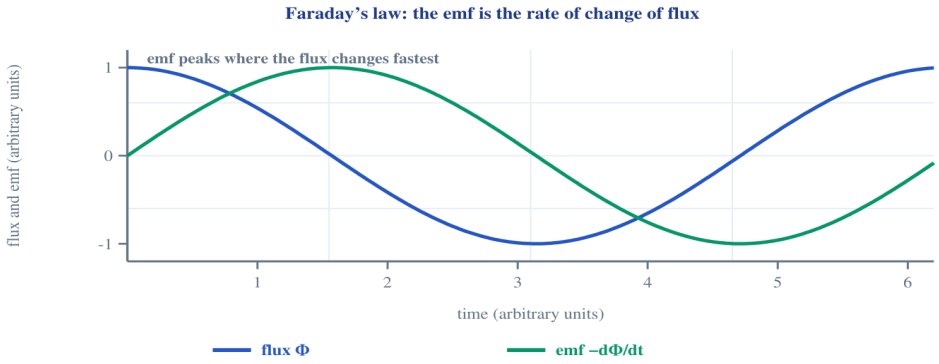
$$(F)/() = (\mu_0\ I_1I_2)/(2\pi\ d). \tag{8}$$

Historically this relation **defined** the ampere. A current loop in a field feels a torque $\tau = m\times B$, where the magnetic moment is m = NI A; this is the electric motor, and it is why a compass needle turns.

5. Electromagnetic induction: Faraday and Lenz

Move a conductor through a field, or change the field through a loop, and a voltage appears. Faraday's law states it exactly:

$$\varepsilon = -(d)/(dt),\ = \int B\cdot\ d\ A = BAcos\ \theta. \tag{9}$$



A loop rotating in a uniform field produces a sinusoidal emf; a coil entering a field experiences a growing then falling flux.

Figure. A loop rotating in a uniform field produces a sinusoidal emf; a coil entering a field experiences a growing then falling flux.

The minus sign is **Lenz's law**: the induced current flows in the direction that **opposes** the change of flux that caused it. If the flux into the page is growing, the induced current circulates so that its own field points out of the page. Lenz's law is energy conservation in disguise — a current that reinforced its own cause would produce energy from nothing.

Three everyday versions of the same law:

- **Motional emf.** A rod of length moving at speed v perpendicular to B acts as a battery with ε = B v. The charges are pushed by the q v× B force until the electric field balances it.
- **Rotating coil.** A coil of N turns and area A spinning at angular frequency ω in a field gives ε = NBAωsin ω t — the alternator, and the reason mains supply is sinusoidal.
- **Eddy currents.** A changing flux in a solid conductor drives loops of current that dissipate energy. That is magnetic braking, the induction hob, and the reason transformers are laminated.

The flux can change because B changes, because the area changes, or because the orientation changes; the exam skill is to identify which of the three is happening before differentiating.

6. Self-inductance and magnetic energy

A circuit's own current creates flux through itself: = LI, with L measured in henries. Any change of current is then opposed by a back-emf

$$\varepsilon = -L(dI)/(dt), \tag{10}$$

so an inductor is **electrical inertia**: the current cannot jump, exactly as a mass's velocity cannot jump. The energy stored in the field is

$$U = (1)/(2)\ LI^2. \tag{11}$$

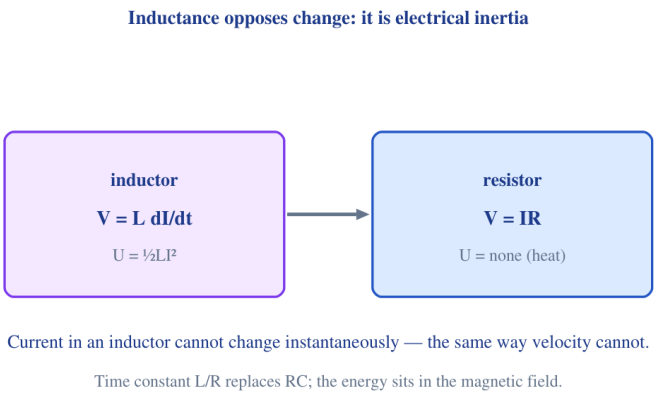


Figure 4. An inductor stores energy in its field and resists current change; a resistor dissipates energy and does not resist change, only flow.

For a long solenoid of N turns, length and area A (so n = N/), L = μ_0 n² A = μ_0 N²A/. Dividing U by the volume gives the magnetic energy density,

$$u_B = (B^2)/(2\mu_0), \tag{12}$$

the exact analogue of (1)/(2)ε_0E² for the electric field — a pairing worth remembering, because it lets you compute the inductance of a field configuration without ever drawing a circuit.

In an RL circuit the current approaches its final value with time constant τ = L/R, just as a capacitor charges with τ = RC. The roles are exchanged: a capacitor resists a change of **voltage**, an inductor resists a change of **current**.

The magnetism checklist. Direction first (right-hand rules) → is the force doing work? (no) → circular motion r = p/(qB) and T = 2π m/(qB) → for induction, identify **what** is changing the flux → apply Lenz to fix the sign → energy: (1)/(2)LI² for a circuit, B²/(2μ_0) per unit volume for the field.

7. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Field near a wire and at the centre of a loop. A long straight wire carries 8.0 {A}. (a) Find the field 5.0 {cm} from the wire. (b) Find the field at the centre of a single circular loop of radius 4.0 {cm} carrying the same current. (c) A solenoid of length 25 {cm} with 400 turns carries 3.0 {A}; find the field inside and compare all three results.

Worked solution →

[1] Problem 2 — A proton in a cyclotron. A proton enters a uniform 0.50 {T} field at 5.0×10^6 {m,s^(-1)}, perpendicular to the field. (a) Find the radius of its path. (b) Find the period and the frequency. (c) Find its kinetic energy in keV and explain why doubling the speed leaves the period unchanged while doubling the field halves it.

Worked solution →

[2] Problem 3 — A velocity selector and a mass spectrometer. A beam of singly charged ions passes undeflected through crossed fields E = 20 {kV,m^(-1)} and B = 0.40 {T}. (a) Find the ion speed. (b) The ions then enter a region with only the magnetic field; find the radius for a potassium-39 ion. (c) Find the mass of an ion whose radius is 4.0 {cm} larger, in atomic mass units, and explain why such an instrument can separate isotopes.

Worked solution →

[2] Problem 4 — Force on a wire and a loop. A straight wire of length 25 {cm} carries 4.0 {A} in a field of 0.30 {T} at right angles. (a) Find the force. (b) Find the maximum torque on a 100-turn rectangular coil 5.0 {cm}× 8.0 {cm} carrying 2.0 {A} in a field of 0.15 {T}. (c) Explain what happens to a loop that is free to rotate, and what limits its rotation in a real motor.

Worked solution →

[2] Problem 5 — Two parallel currents. Two long parallel wires 4.0 {cm} apart carry 15 {A} and 20 {A} in the same direction. (a) Find the force per unit length on each and state whether it is attractive or repulsive. (b) Find the point between them where the net field is zero. (c) Find the force if the second current is reversed and describe the resulting motion.

Worked solution →

[2] Problem 6 — A rod on rails (motional emf). A rod of length 40 {cm} slides at 3.0 {m,s^(-1)} on frictionless rails in a field of 0.50 {T} perpendicular to the plane. The circuit is closed by a resistor of 0.60 Ω; the rod's resistance is negligible. (a) Find the emf, the current and the force needed to keep the rod moving. (b) Find the power delivered mechanically and the power dissipated in the resistor. (c) Show that the magnetic force on the rod always opposes its motion, and explain what happens to the energy if the rod is released.

Worked solution →

[2] Problem 7 — A rotating coil generator. A coil of 200 turns and area 0.12 {m²} rotates at 3000 revolutions per minute in a field of 0.020 {T}. (a) Find the peak emf. (b) Find the rms emf. (c) Find the number of turns needed to obtain an rms output of 230 {V} at the same speed and field.

Worked solution →

[3] Problem 8 — A solenoid as an inductor. A solenoid 20 {cm} long and 1.2 {cm} in radius has 1200 turns. (a) Find its inductance. (b) Find the energy stored when the current is 2.5 {A}, and the energy density in the field. (c) Find the time constant with a 0.50 Ω resistor and explain what happens at the instant the switch is closed.

Worked solution →

[3] Problem 9 — A changing field through a loop. A circular loop of radius 5.0 {cm} and resistance 0.020 Ω lies in the plane of the page. A field perpendicular to the page increases at 0.40 {T,s^(-1)}. (a) Find the induced emf and current. (b) State the direction of the current and justify it with Lenz's law. (c) Find the power dissipated and compare it with the rate of change of field energy, explaining where the energy comes from.

Worked solution →

[3] Problem 10 — Helical motion and magnetic confinement. A proton enters a field of 0.25 {T} at 4.0×10^6 {m,s^(-1)}, at 30° to the field lines. (a) Find the radius and period of the circular part of its motion. (b) Find the pitch of the helix. (c) A field that is stronger at its ends turns the particle around: explain, using r = p/(qB) and energy conservation, why this "magnetic mirror" works and where the particle's axial momentum goes.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

> **MIT OpenCourseWare 8.02, Electricity and Magnetism**
> Full lecture notes and problem sets for fields, circuits and induction.